

# Filo: Empowering young learners at scale with instant online tutoring

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## ABOUT FILO

Filo executed a c  
Google Cloud mig  
with zero downtir  
its ed-tech platfo  
million users in 2  
and forged an AI-  
global expansion  
with Google Clou  
Google Workspac



## About Filo

Filo is a US-based online education platform with a presence in 15 countries around the world. It connects any student with a tutor within 60 seconds. Under a philosophy of being "every student's companion on good and bad days of their learning journey," Filo has gained more than four million users across India and beyond since launching in 2020. Inspired by diverse learning styles of individual students, Filo strives to connect young learners with tutors who can foster personal growth in each education experience. The startup is a recipient of the

## Google Cloud results

- Drives ed-tech empowerment mission by seamlessly scaling to 4 million users in 2 years
- Promotes student satisfaction by executing complex migration with no platform downtime
- Enables investment in app development by reducing cloud computing costs by 60% through GKE spot instances
- Creates safe and secure environment for young learners

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Google Play Users' Choice App Award 2022 in the category "Best for Personal Growth" and the International Glory Awards 2022 "Best Learning Platform" award.

Industries: Technology

Location: India

CDN (<https://cloud.google.com/cdn/>). Cloud

# searce°

## About Searce

Searce is one of the largest Google Cloud partners globally with over 12 years of experience, 150+ Google Cloud certifications, and 300+ Google Cloud Experts in-house. It works with digital native companies and enterprises to re-imagine and futurify their business.

Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud CDN (<https://cloud.google.com/cdn/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Kubernetes Engine  
(<https://cloud.google.com/kubernetes-engine/>)

through AI  
content  
filtering tools

- Frees up  
operational  
and budgetary  
resources  
through  
integration of  
Google  
Workspace  
and Google  
Apps Script

Vision AI (<https://cloud.google.com/vision/>) ud

partners globally with over 12 years of

Google Workspace

(<https://workspace.google.com/>)

Sheets

(<https://workspace.google.com/products/sheets/>)

Apps Script

(<https://workspace.google.com/products/apps-script/>)

Tutoring is one of the most empowering methods of helping young learners unlock personal growth to get the most from their education. According to the Brookings Institution

(<https://www.brookings.edu/articles/tutoring-a-time-tested-solution-to-an-unprecedented-pandemic/>)

, studies consistently show that students obtain more than a 15 point percentile gain in test scores with tutoring. However, access to quality tutoring is often unavailable to students when they need it most.

Registered under a parent company in the US, Filo (<https://askfilo.com/en-us>) is closing the gap with an online platform that connects a student to a tutor for a live one-on-one session within 60 seconds. The app aims to be "every student's companion on good days and bad days of their learning journey." It offers real-time attention tailored to the diverse

personalities and learning styles of individual students.

Established in 2020, Filo initially built its solution with a low-budget on a local cloud provider that met its initial operational needs. However, as the platform grew in popularity and expanded to regions across India, Filo found itself struggling with latency and scalability issues. The original vendor lacked data centers with the compute power and agility to service the remote areas where Filo was serving its students.

"Choosing  
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Workspace  
exceeded our  
expectations.  
Not only did it  
bring seamless  
autoscaling and  
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latencies, but  
their intuitive  
use and fully-  
managed

services,  
including out-of-  
box artificial  
intelligence  
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—Shadman Anwer, Co-  
founder & CTO, Filo

Deciding it needed a leading cloud provider, Filo embarked on a full-fledged migration to Google Cloud (<https://cloud.google.com/>) due to the seamless scalability it could achieve with Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>) (GKE) for its microservices-based architecture. The startup was also drawn to how Cloud CDN (<https://cloud.google.com/cdn>) would minimize latency for real-time video tutoring thanks to Google's global edge network. Lastly, Filo saw limitless potential to improve its teaching with Google Cloud's fully managed artificial intelligence tools, such as Vision AI (<https://cloud.google.com/vision>).

Migrating to Google Cloud with Google partner Searce, Filo was able to unleash the power of its ed-tech solution for unlimited scale with stable video connectivity, no matter what the spikes in demand.

This was achieved with a small team of just five engineers, thanks to the fully automated managed services of Google Cloud and proactive partner support. The adoption of Google Workspace (<https://workspace.google.com/>), meanwhile, drove significant cost and operational efficiencies thanks to integration of Sheets and Google Apps Script.

"We realized early on that we needed to move on to a world-class platform that could provide our fast-growing customer base with better connectivity," says Shadman Anwer, Co-founder & CTO, Filo.

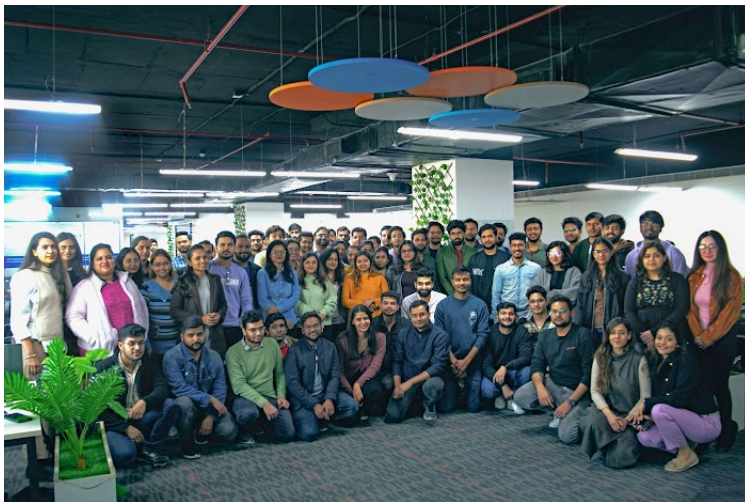
"Choosing Google Cloud and Google Workspace exceeded our expectations. Not only did it bring seamless autoscaling and near-zero latencies, but their intuitive use and fully-managed services, including out-of-box artificial intelligence solutions, liberated our team to drive creative ed-tech empowerment."

## **Drive unlimited growth and cost efficiencies with autoscaling**

For a startup with a lean team, Anwer says the automated environment of GKE to power the platform proved to be a game-changer in the company's journey to reach students across India and beyond. In particular, he says GKE offered optimal versatility for diverse compute workloads that enables Filo to achieve significant cost efficiencies, which are crucial to its success as a startup.

Running its entire backend on GKE, Filo deploys numerous GPU and non GPU node pools for wide-

ranging computing needs. For example, it runs artificial intelligence (AI) and machine learning (ML) pipelines on the most powerful GPU node pools, while deploying standard API services that connect to the frontend with non GPU node pools. It then uses affordable GKE spot instances for basic infrastructure tasks that are not mission critical and can run at a delay, saving around 60% in cloud computing costs. In all operational modes, says Anwer, GKE ensured that Filo enjoyed the high availability needed to serve learners around the clock with zero interruptions.



*Filo Team*

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"Google Cloud  
provides us a  
huge global

infrastructure,  
and also a deep  
local  
infrastructure.  
It's the best of  
both worlds.  
This allowed us  
to not only  
expand globally  
very efficiently,  
but also tap  
customers deep  
within India  
itself, and help  
us provide them  
a wonderful and  
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experience."

—Shadman Anwer, Co-  
founder & CTO, Filo

The results Filo has achieved through its multi-tier GKE-based architecture have been impressive. The startup has scaled seamlessly to more than four million users with a range of fully-managed Google Cloud services, including Cloud SQL as its relational database and BigQuery.

(<https://cloud.google.com/bigquery>) as its data



warehouse. Leveraging Cloud SQL (<https://cloud.google.com/sql>), Filo's data storage scaled easily from 100 gigabytes to close to nearly 1.5 terabytes, without the DevOps team needing to lift a finger on database configurations.

"Google Cloud provides us a huge global infrastructure, and also a deep local infrastructure. It's the best of both worlds. This allowed us to not only expand globally very efficiently, but also reach out to customers across India itself, helping us provide them a wonderful and swift learning experience," says Anwer. "Meanwhile, GKE spot instances let us build cost-optimized, high-compute workflows such as video pipelines, allowing us to be frugal and generate better cost margins in our business."

## **Enhancing the tutoring experience while nurturing a safe environment**

Filo found Vision AI to be an essential tool in driving optical character recognition (OCR) for a variety of educational needs, such as monitoring handwriting and creating content filters. Leveraging the AI solution as a powerful, out-of-box solution gave Filo significant versatility and flexibility to integrate ML enhancements into its platform, improving the tutoring experience while making it safe and secure for young learners.

"Vision AI was a major enabler for us. If we needed to manually train models for OCR, it would be a huge drain on resources. With these managed APIs, we were able to achieve so much with fewer resources in a shorter amount of time."

*—Umesh Kumar Sharma,  
Head of Engineering, Filo*

"Vision AI was a major enabler for us. If we needed to manually train models for OCR, it would be a huge drain on resources," says Umesh Kumar Sharma, Head of Engineering, Filo. "With these managed APIs, we were able to achieve so much with fewer resources in a shorter amount of time. That's been one of the main benefits of choosing Google Cloud."

## Optimizing operational processes via automation

Another powerful way that Filo is able to achieve outcomes at scale with a lean team is its creative deployment of Google Apps Script

(<https://www.google.com/script/start/>), which helps Filo create business applications that integrate with Google Workspace quickly and easily. Filo has been driving operational automations, such as user fees and payments to third-party vendors, by creating simple programs on Apps Script that run on Sheets (<https://www.google.com/sheets/about/>).

The automation enabled by the seamless integration between Apps Script and Sheets means Filo need not run separate customer relations management (CRM) software, or expend back office staff time, to handle interactions, such as user registration, authentication, and payments.

"The automations enabled by Apps Script and Sheets have been very important for us. They are the most affordable tools for automating operational processes without investing in CRMs," he says. "By avoiding the need to invest in and test various tools, we've been able to save both time and money."

## Combining AI with the human factor for empowering students

Looking into the future, Filo plans to further enhance its platform with more sophisticated AI-enabled solutions. These will deploy large language models

(LLMs) that complement the tutoring performed by teachers so as to maximize the students' potential.

Filo is focused on leveraging AI to help tutors become better at what they do, such as giving them instant access to scientific formulas they might not have at their fingertips. To drive this vision, Filo plans to deploy Vertex AI

(<https://cloud.google.com/vertex-ai>) to train LLMs that enable the seamless integration of human mentorship and AI-enabled pedagogy.

The startup also plans to leverage Vertex AI to train LLMs to read transcripts of a session and judge the quality of tutoring sessions. Filo has 40 different parameters to assess teaching quality, ranging from whether the teacher has created a comfortable learning session, to the time required to solve a student's problem. LLMs built with Vertex AI can thus automate much of the quality control for Filo, thus further enhancing the startup's mission while driving cost and operational efficiencies.

"We see an exciting future at the intersection of human and AI-driven teaching in which Vertex AI can play a vital role," says Anwer. "The combination of a human touch and personalized learning will combine with powerful AI insights to drive endless cycles for student empowerment far into the future."